Ingénierie des circuits intégrés

INTRODUCTION

Plan

Physique des composants et technologies des capteurs

ü Phénomènes de transports dans les semiconducteurs

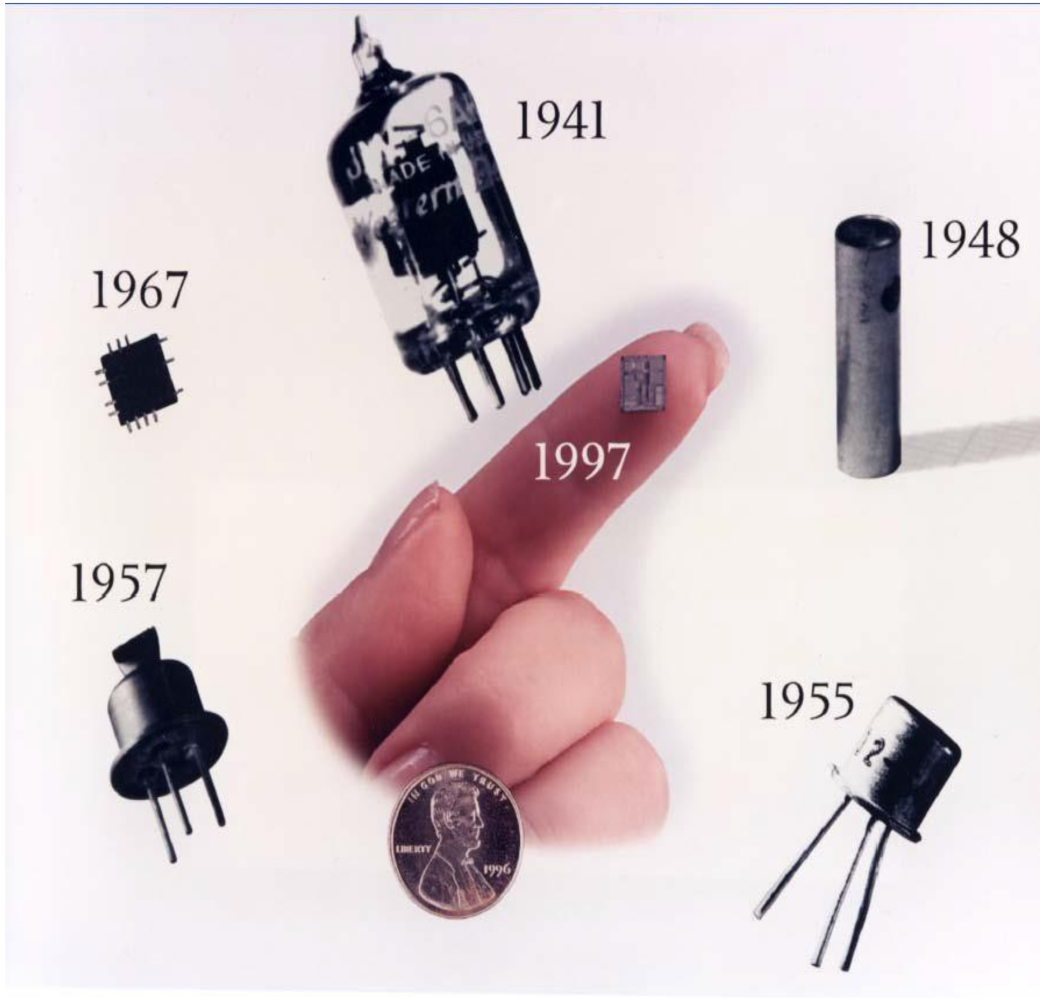
ü Jonction : Jonction PN et Jonction métal semiconducteur

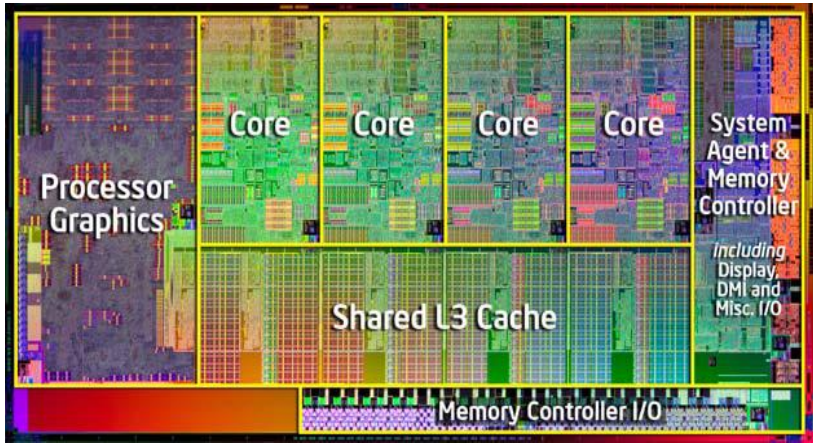
ü Structure MOS, transistor MOS

ü Transistors : Bipolaire et MOS

Introduction à la Conception de circuits intégrés

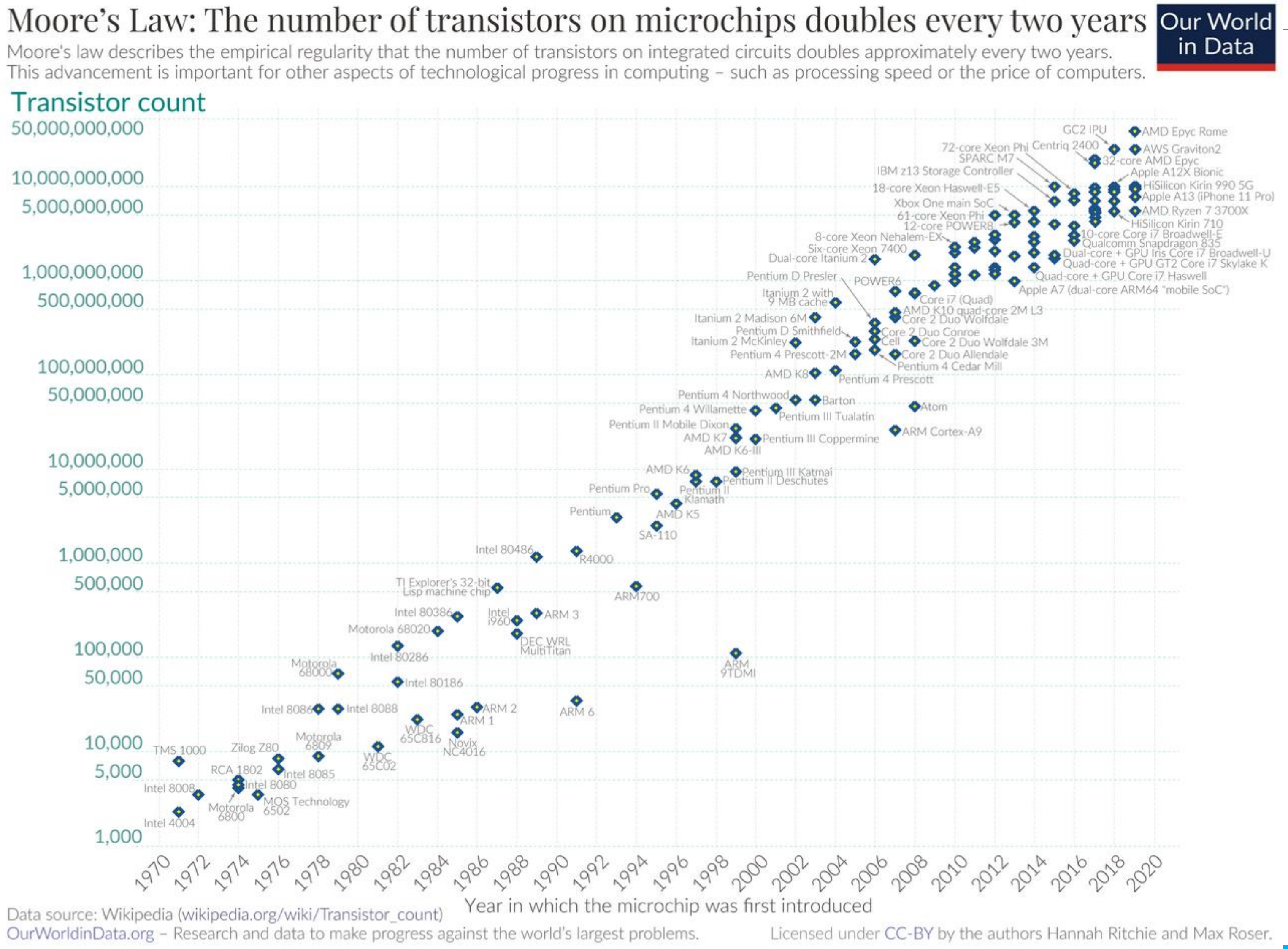
Évolution historique





2011 : Intel Core I7 2600K, 32nm :
Die : 216mm^2
1,16 x 10^9 transistors !

Évolution historique



MOSFET scaling (process nodes)

- 20 µm – 1968
- 10 µm – 1971
- 6 µm – 1974
- 3 µm – 1977
- 1.5 µm – 1981
- 1 µm – 1984
- 800 nm – 1987
- 600 nm – 1990
- 350 nm – 1993
- 250 nm – 1996
- 180 nm – 1999
- 130 nm – 2001
- 90 nm – 2003
- 65 nm – 2005
- 45 nm – 2007
- 32 nm – 2009
- 28 nm – 2010
- 22 nm – 2012
- 14 nm – 2014
- 10 nm – 2016
- 7 nm – 2018
- 5 nm – 2020
- 3 nm – 2022

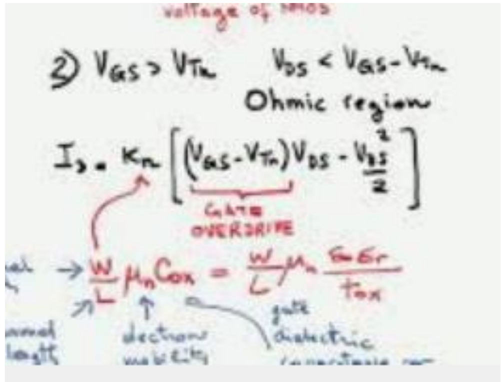
Future

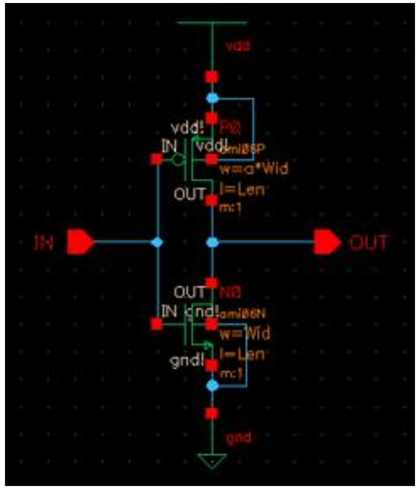
- 2 nm ~ 2025
- 1 nm ~ 2027

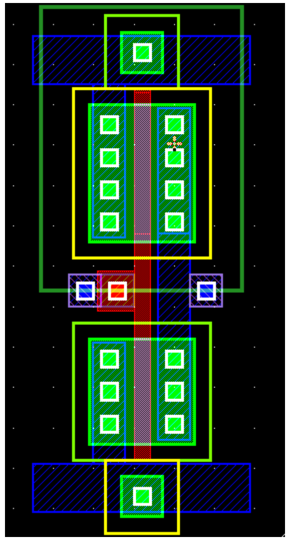
Évolution historique

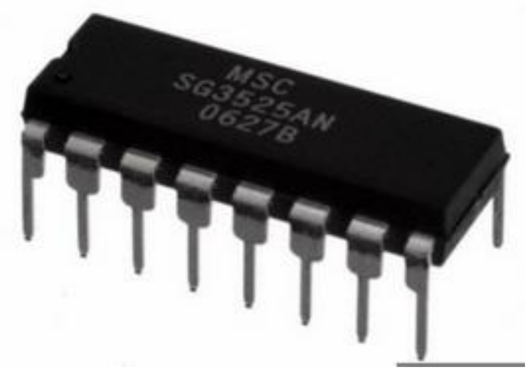
- 1960-2025 Réduction des tailles de transistors
  - Évolution anticipée (loi de Moore)
  - Industrie au cœur de la croissance économique des 60 dernières années
  - Augmentation exponentielle des performances
    - ✓ Densité (Moore) : nbre de Trans. Double tous les 18 mois
    - ✓ Transistors par puce = $2^{\text{année}-1964}$

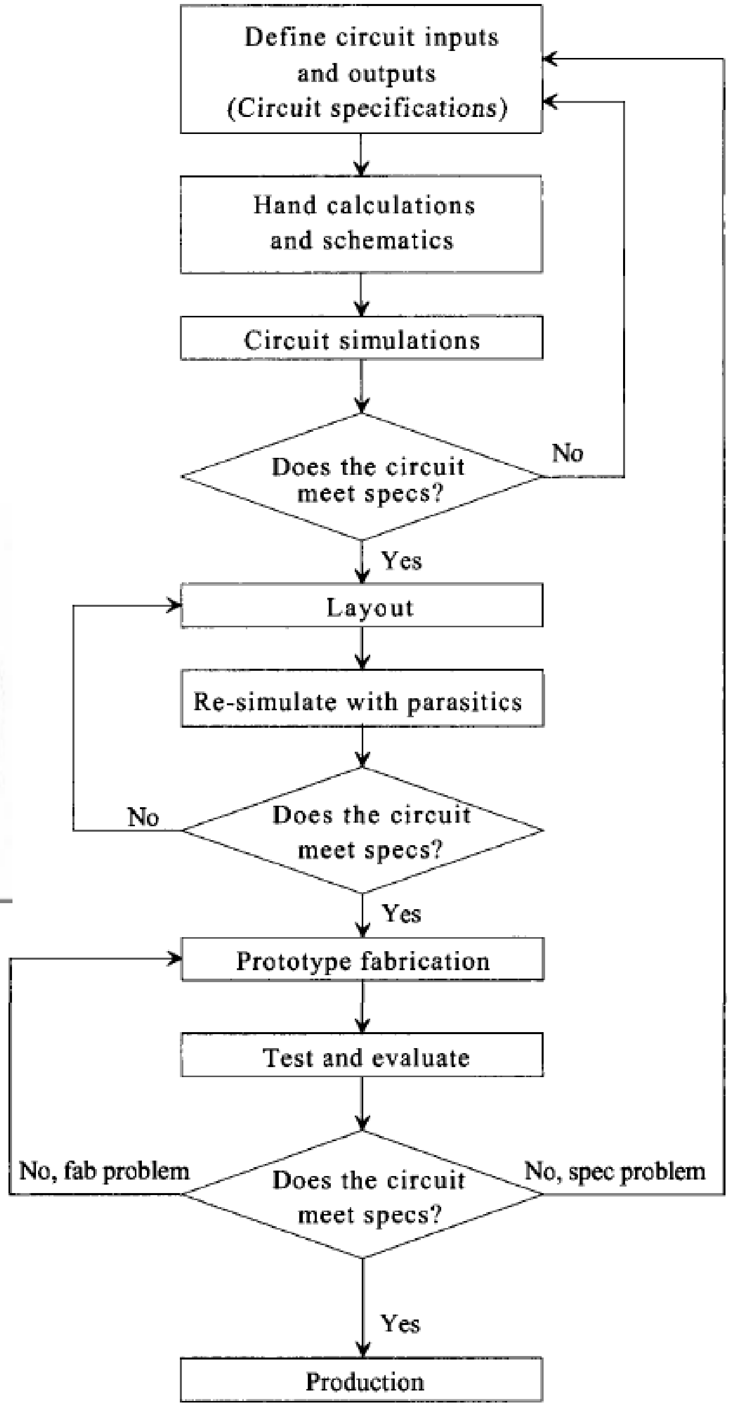
Phases de conception des circuits intégrés (CI)



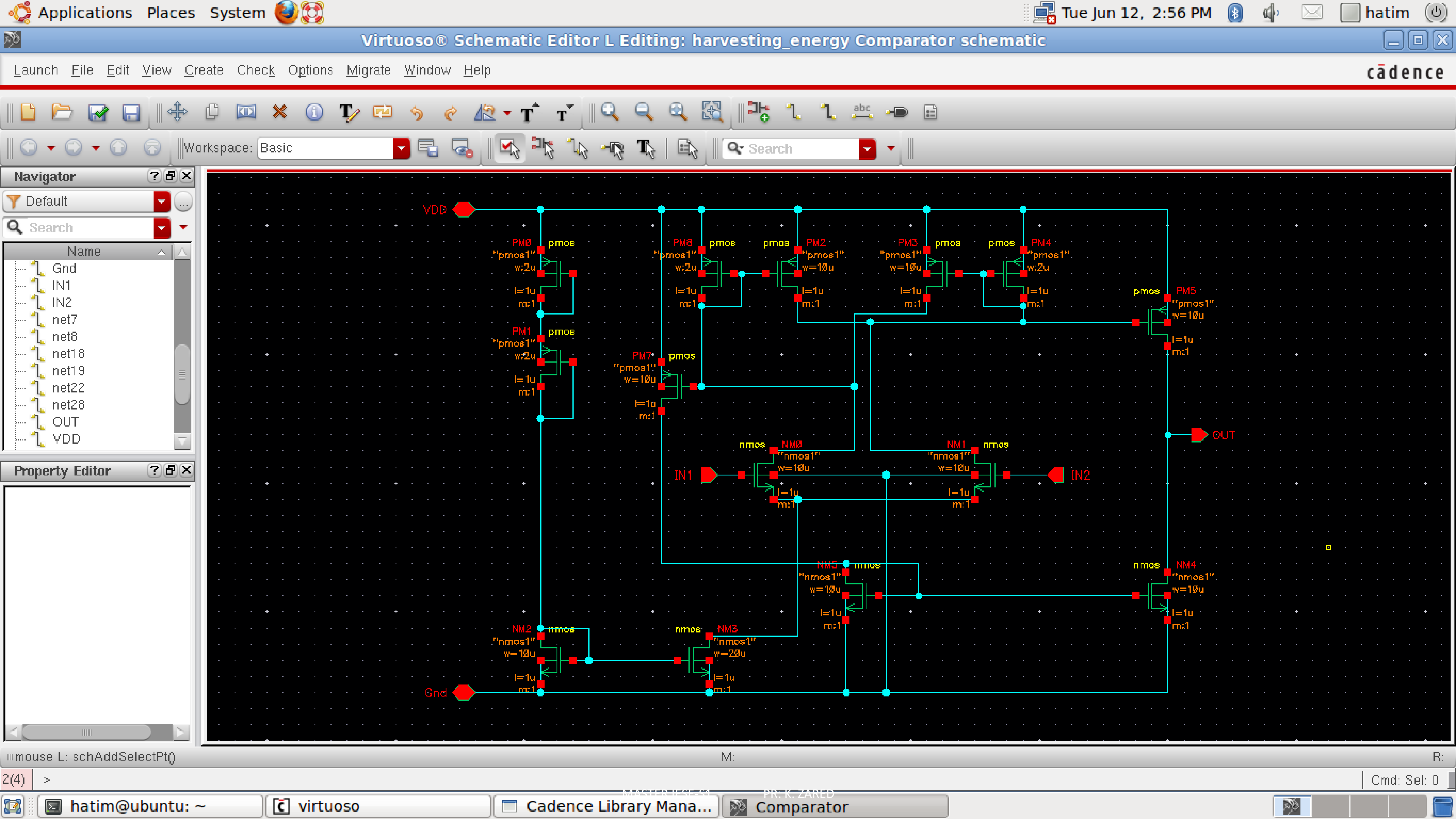








CMOS Circuit Design, Layout and Simulation. R. JACOB BAKER, IEEE Series on Microelectronic Systems, Third Edition, 2010.



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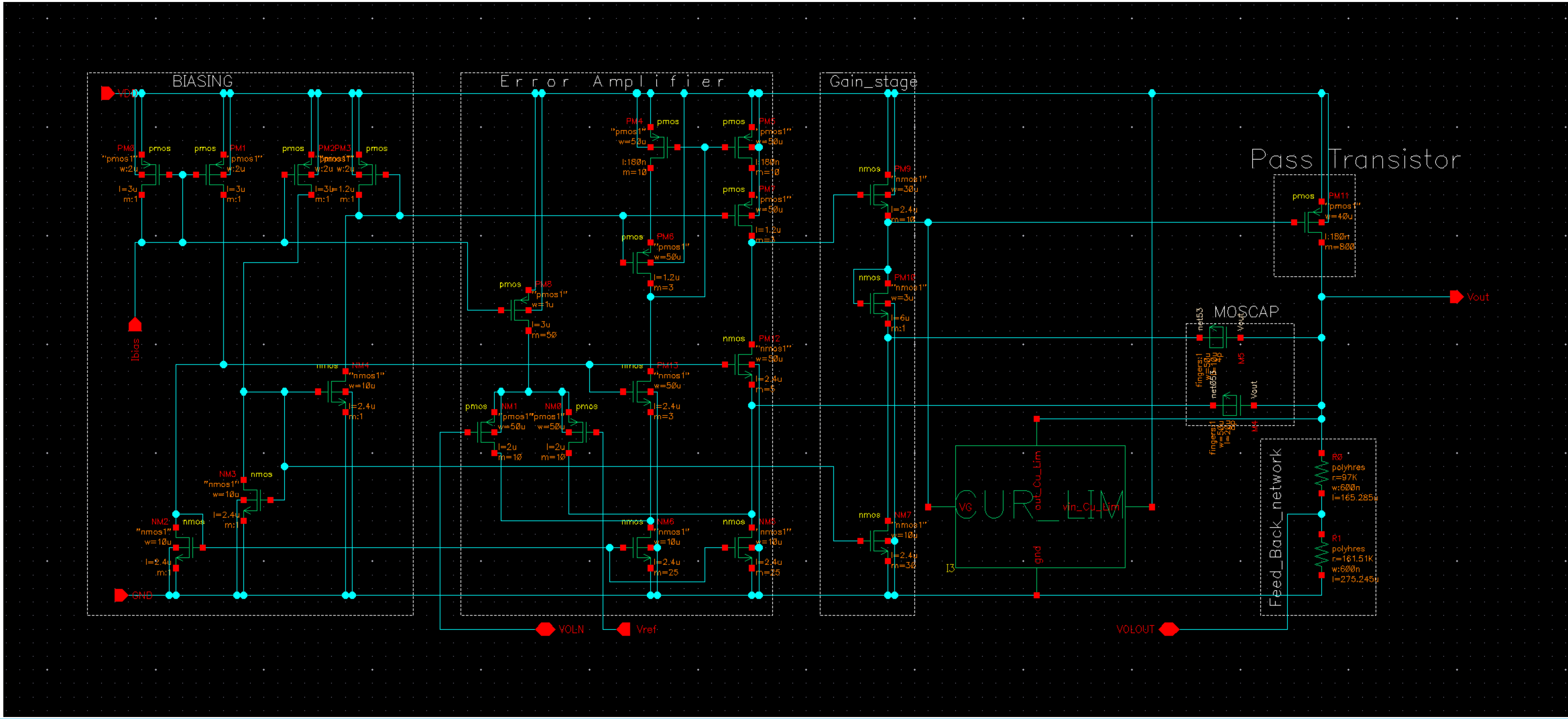
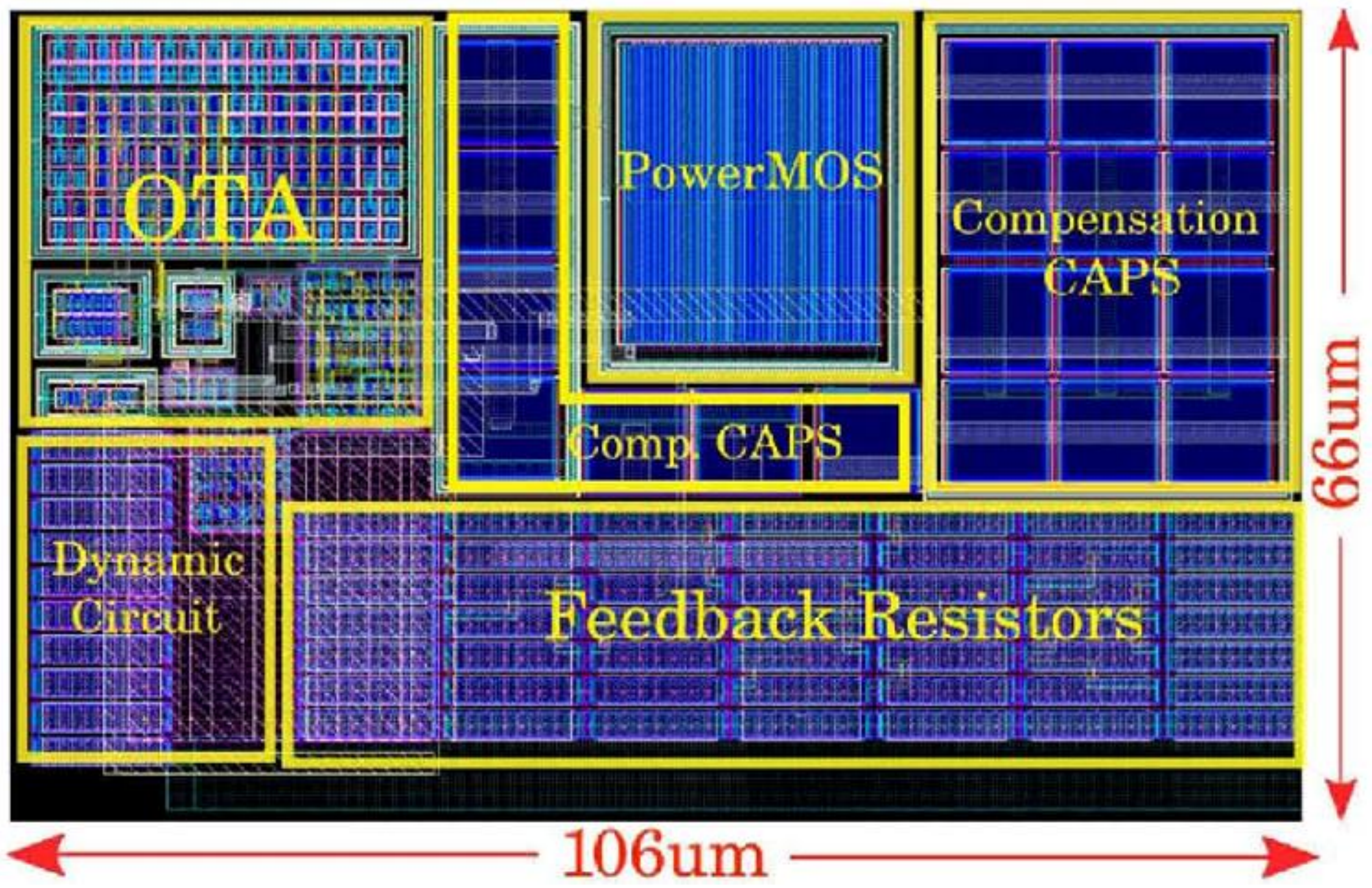
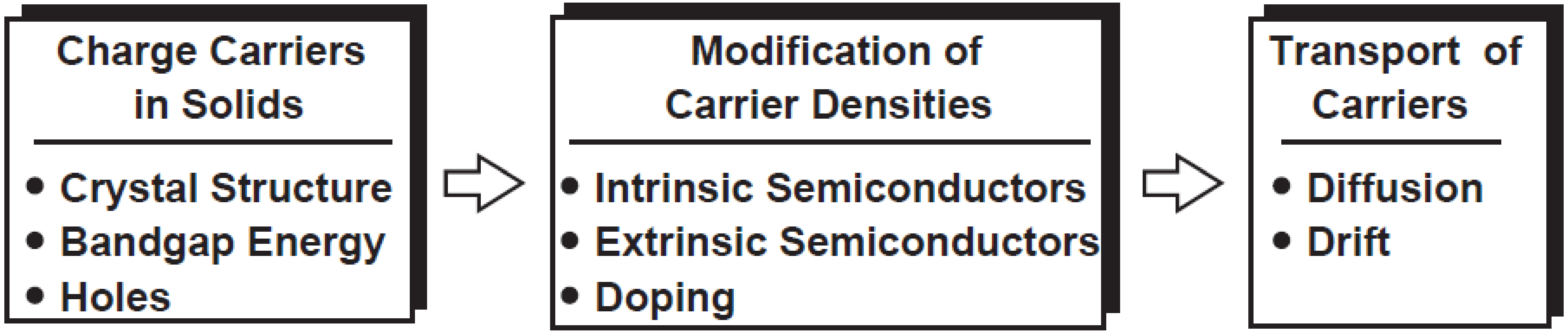
Phases de conception des circuits intégrés (CI)



Phases de conception des circuits intégrés (CI)

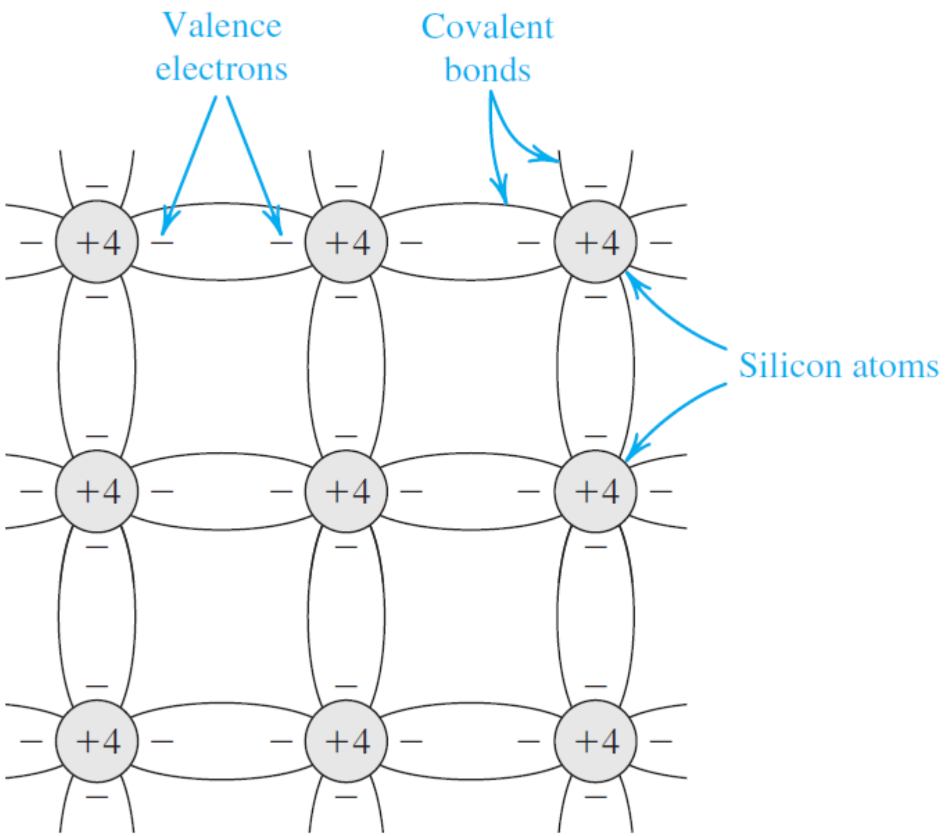


Physics of Semi-Conductors



Physics of Semi-Conductors/Intrinsic Semiconductors

- **Charge Carriers in Solids**



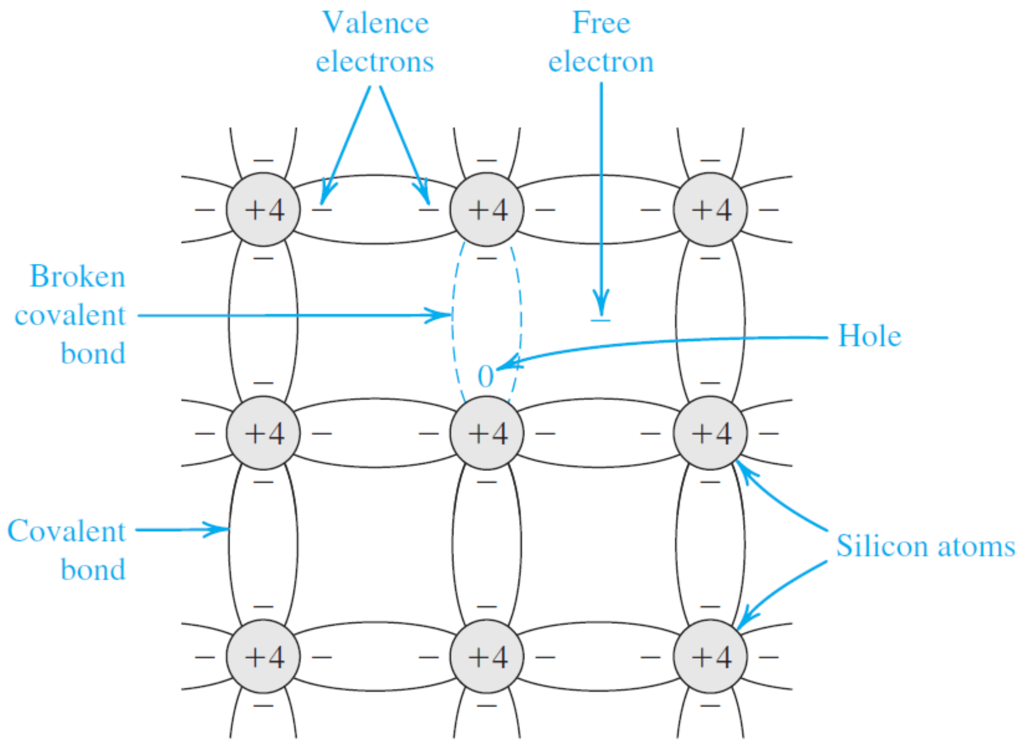
Two-dimensional representation of the silicon crystal

conductivity lies between that of conductors, such as copper, and insulators, such as glass

III	IV	V
Boron (B)	Carbon (C)	
Aluminum (Al)	Silicon (Si)	Phosphorus (P)
Galium (Ga)	Germanium (Ge)	Arsenic (As)

Physics of Semi-Conductors

- **Charge Carriers in Solids**

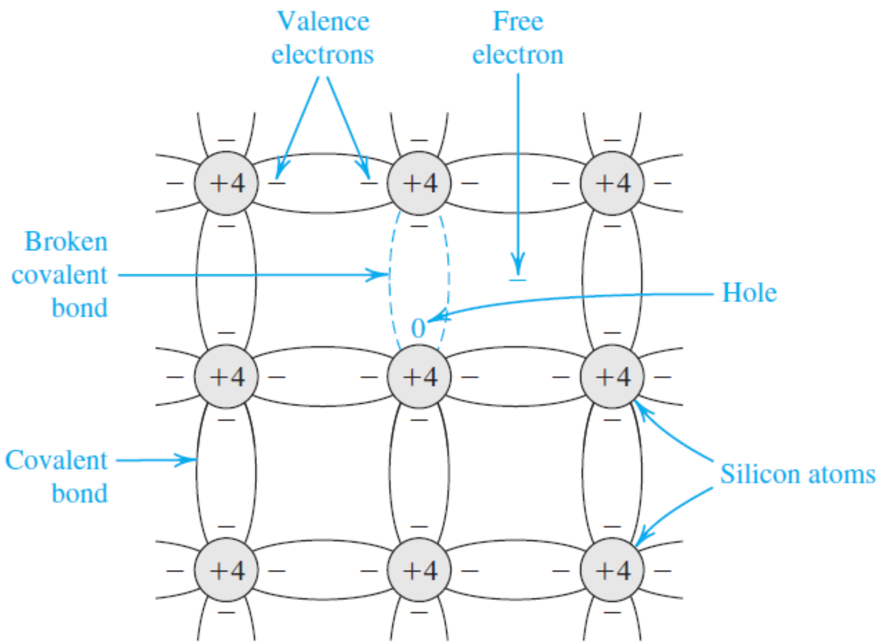


At room temperature, some of the covalent bonds are broken by thermal generation

A silicon atom residing in isolation contains four valence electrons requiring another four to complete its outermost shell. If processed properly, the silicon material can form a "crystal" wherein each atom is surrounded by exactly four others. As a result, each atom *shares* one valence electron with its neighbors, thereby completing its own shell and those of the neighbors. The "bond" thus formed between atoms is called a "covalent bond" to emphasize the sharing of valence electrons.

Physics of Semi-Conductors

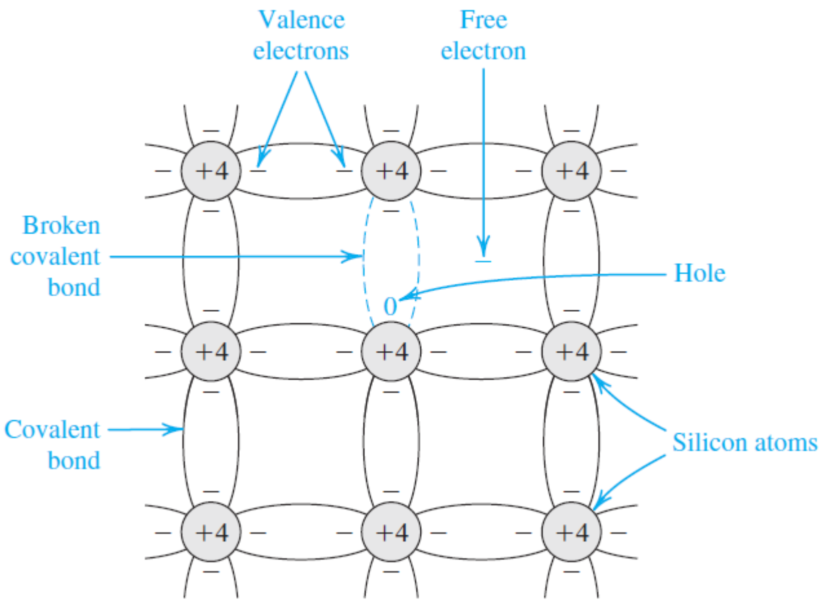
- **Charge Carriers in Solids**



- At temperatures near absolute zero, the valence electrons are confined to their respective covalent bonds, refusing to move freely.
- The silicon crystal behaves as an insulator for $T \to 0K$.
- At higher temperatures, electrons gain thermal energy, occasionally breaking away from the bonds and acting as free charge carriers until they fall into another incomplete bond (free electrons).
- Thermal generation results in free electrons and holes in equal numbers and hence equal concentrations (the number of charge carriers per unit volume (cm^3).
- The free electrons and holes move randomly through the silicon crystal structure, and in the process some electrons may fill some of the holes.

Physics of Semi-Conductors

- **Charge Carriers in Solids**



- In thermal equilibrium, the recombination rate is equal to the generation rate, and can conclude that the concentration of free electrons n is equal to the concentration of holes p,

$$n = p = n_i \quad (1)$$

where n_i denotes the number of free electrons and holes in a unit volume (cm^3) of intrinsic silicon at a given temperature. from semiconductor physics gives *ni* as

$$n_i = BT^{3/2} e^{-E_g/2kT}$$

where B is a material-dependent parameter that is $7.3\times10^{15} cm^{-3}K^{-3/2}$ for silicon; T is the temperature in K; Eg, a parameter known as the **bandgap energy**, is 1.12 electron volt (eV) for silicon; and k is Boltzmann's constant (8.62×10^{-5} eV/K).

With $1eV = 1.6\times10^{-19}$ J

Physics of Semi-Conductors

Example 1

Calculate the value of n_i for silicon at room temperature ($T \simeq 300$ K).

Solution

Substituting the values given above in Eq. (2) provides ;

$$n_i = 7.3 \times 10^{15} (300)^{3/2} e^{-1.12/\left(2 \times 8.62 \times 10^{-5} \times 300\right)}$$

$$= 1.5 \times 10^{10} \text{carriers/cm}^3$$

Although this number seems large, to place it into context note that silicon has 5×10^{22} atoms/cm^3.

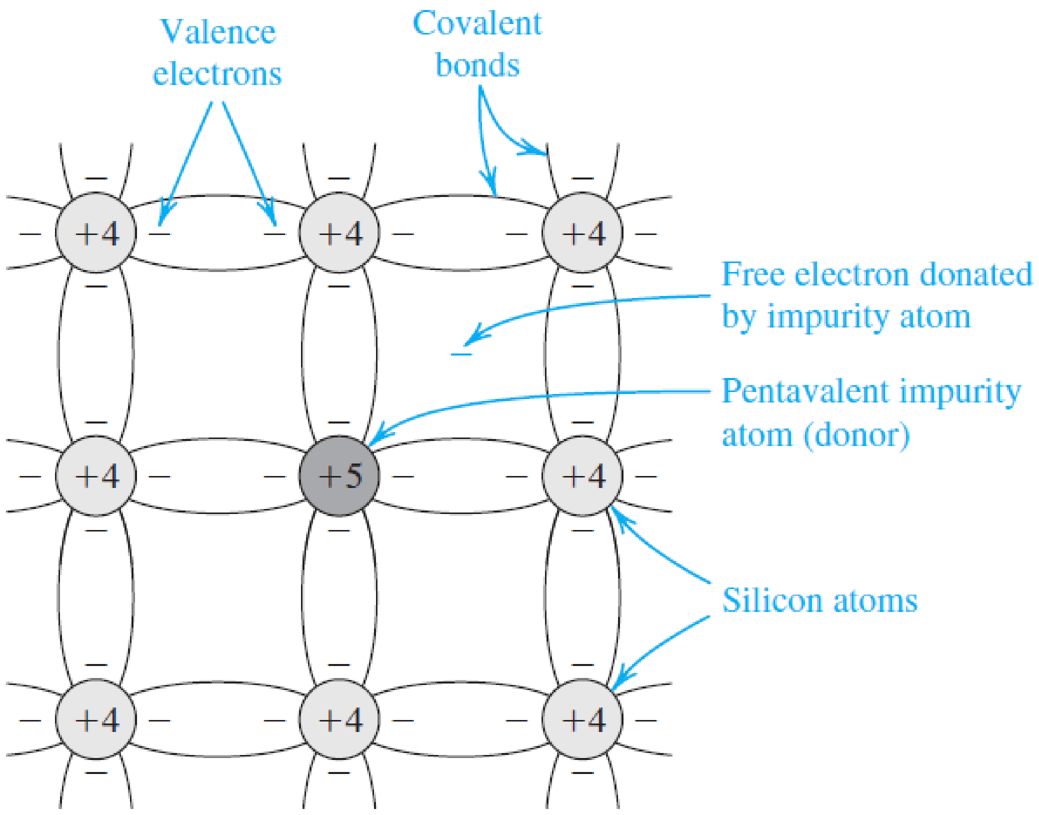
Physics of Semi-Conductors

Doped Semiconductors /n-type

- The intrinsic silicon crystal described above has equal concentrations of free electrons and holes, generated by thermal generation.
- These concentrations are far too small for silicon to conduct appreciable current at room temperature.
- Change the carrier concentration in a semiconductor crystal substantially is the doping process
- Doping involves introducing impurity atoms into the silicon crystal in sufficient numbers to substantially increase the concentration of either free electrons or holes but with little or no change in the crystal properties of silicon.
- To increase the concentration of free electrons, n, silicon is doped with an element with a valence of 5, such as phosphorus.
- The resulting doped silicon is then said to be of ***n* type**.
- To increase the concentration of holes, *p*, silicon is doped with an element having a valence of 3, such as boron, and the resulting doped silicon is said to be of ***p* type**.

Physics of Semi-Conductors

Doped Semiconductors/ n-type



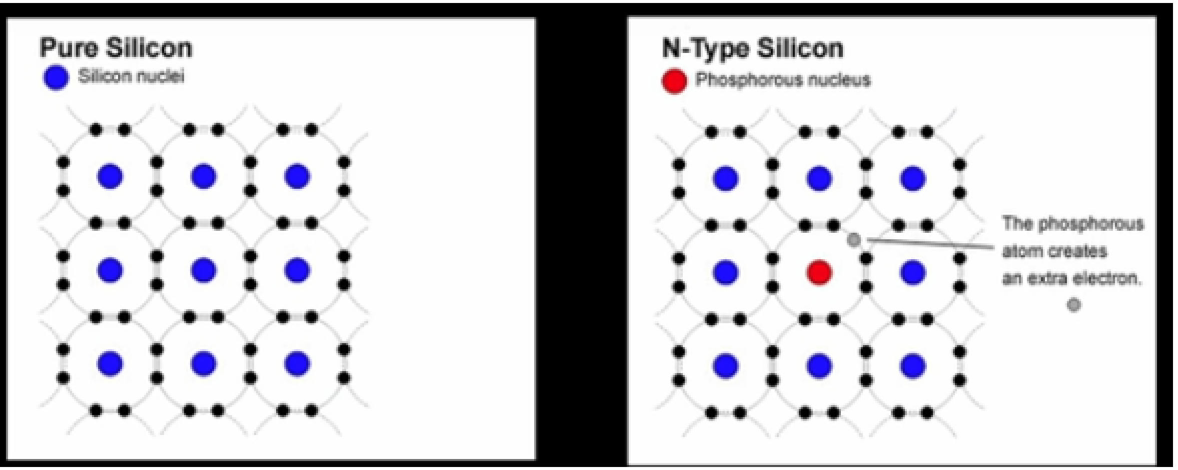
silicon crystal doped with phosphorus impurity

- The dopant (phosphorus) atoms replace some of the silicon atoms in the crystal structure.
- phosphorus atom has five electrons in its outer shell, four of these electrons form covalent bonds with the neighboring atoms, and the fifth electron becomes a free electron.
- Each phosphorus atom *donates* a free electron to the silicon crystal, and the phosphorus impurity is called a **donor**.
- If the concentration of donor atoms is N_D, where N_D is usually much greater than n_i, the concentration of free electrons in the *n*-type silicon will be $\boldsymbol{n_n = N_D}$ ***(3)***

Each dopant atom donates a free electron and is thus called a donor. The doped semiconductor becomes *n* type.

Physics of Semi-Conductors

Doped Semiconductors/ n-type



silicon crystal doped with phosphorus impurity

- The concentration of free electron n_n is determined by the doping concentration and not by temperature.
- All the holes (p_n) in the n-type silicon are those generated by thermal ionization.
- *The* p_n concentration can be found by noting that the relationship (3) at thermal equilibrium by

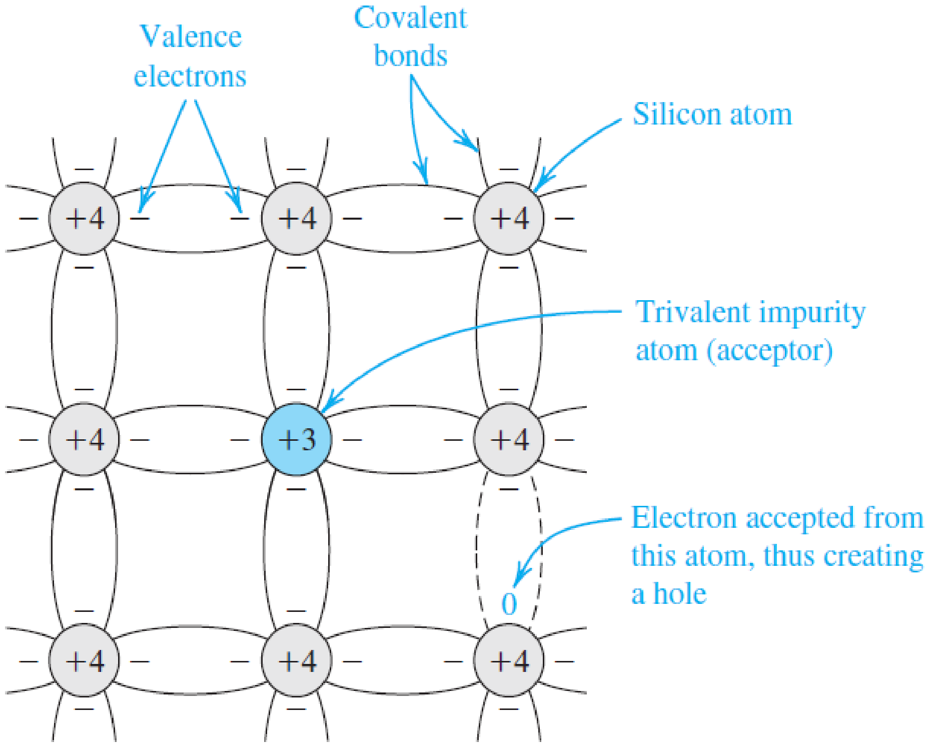
$$p_n n_n = n_i^2$$

$$p_n \simeq \frac{n_i^2}{N_D} \qquad (5)$$

- electrons are said to be the **majority** charge carriers and holes the **minority** charge carriers in *n*-type silicon.

Physics of Semi-Conductors

Doped Semiconductors/ p-type



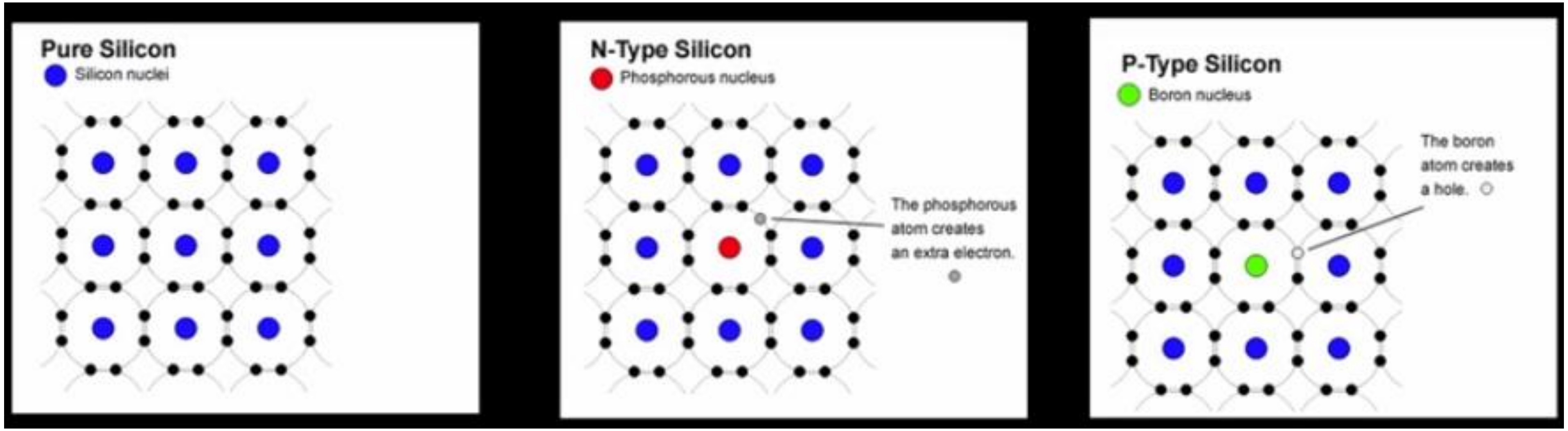
A silicon crystal doped with boron, a trivalent impurity.

- To obtain p-type silicon in which holes are the majority charge carriers, a trivalent impurity such as boron is used.
- each boron atom has three electrons in its outer shell, it accepts an electron from a neighboring atom, thus forming covalent bonds.
- All the holes (p_n) in the n-type silicon are those generated by thermal ionization.
- The result is a hole in the neighboring atom and a bound negative charge at the acceptor (boron) atom.
- each acceptor atom provides a hole.
- If the acceptor doping concentration is NA, where NA>> n_i, the hole concentration becomes $p_p \simeq N_A$ (6)

  where the subscript *p* denotes *p*-type silicon

Physics of Semi-Conductors

- **Doped Semiconductors/ n-type and p-type**



Physics of Semi-Conductors

Example 2

Consider an n-type silicon for which the dopant concentration $N_D = 10^{17}/\text{cm}^3$. Find the electron and hole concentrations at $T = 300$ K.

Solution

The concentration of the majority electrons is

$$n_n \simeq N_D = 10^{17}/\text{cm}^3$$

The concentration of the minority holes is

$$p_n \simeq \frac{n_i^2}{N_D}$$

In Example 3.1 we found that at $T = 300$ K, $n_i = 1.5 \times 10^{10}/\text{cm}^3$. Thus,

$$p_n = \frac{\left(1.5 \times 10^{10}\right)^2}{10^{17}}$$

$$= 2.25 \times 10^3/\text{cm}^3$$

Observe that $n_n \gg p_n$ and that n_n is vastly higher than n_i

Current Flow in Semiconductors

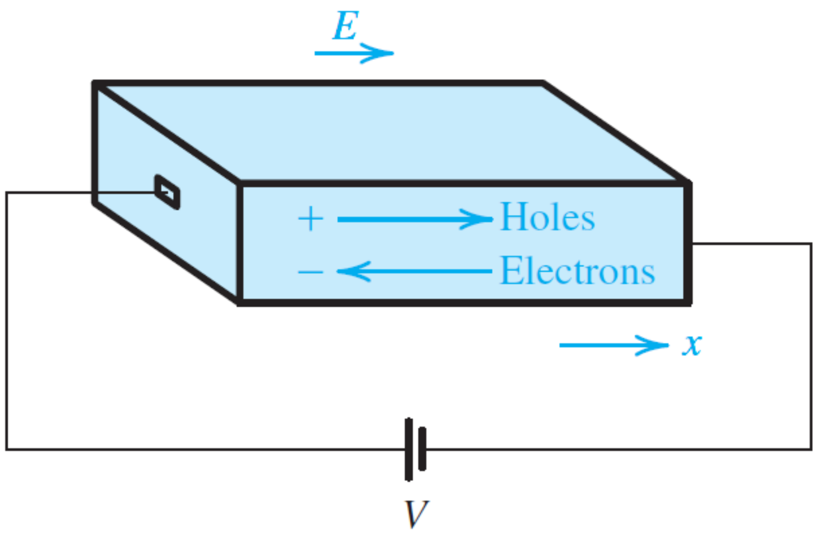
There are two distinctly different mechanisms for the movement of charge carriers and hence for current flow in semiconductors: drift and diffusion.

- When an electrical field *E* is established in a semiconductor crystal, holes are accelerated in the direction of *E*, and free electrons are accelerated in the direction opposite to that of *E*.
- The holes acquire a velocity $v_{p\text{-drift}}$ given by

$$v_{p\text{-drift}} = \mu_p E \qquad (6)$$

where μ_p is a constant called the **hole mobility** with value μp = 480 cm²/V.s.

Drift Current



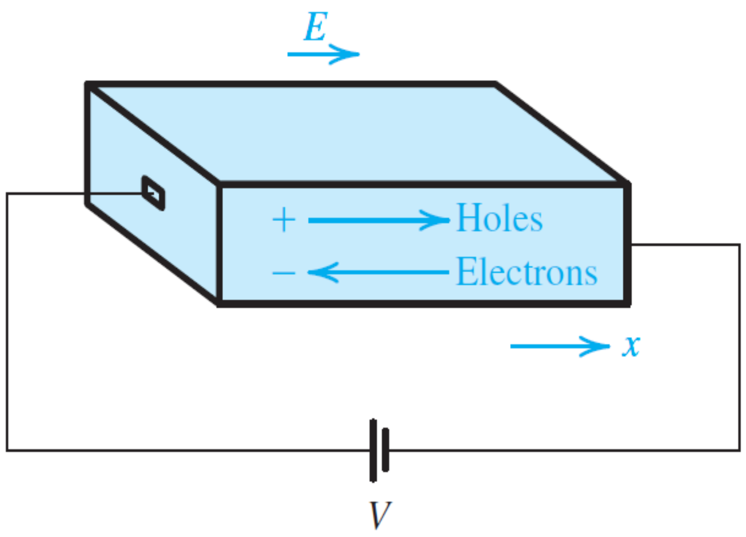
- The free electrons acquire a drift velocity $V_{n\text{-drift}}$ given by

$$v_{n\text{-drift}} = -\mu_n E$$

μ_n is the **electron mobility**, which for intrinsic silicon is about 1350cm²/V

Current Flow in Semiconductors

Drift Current



- to calculate the current component due to the flow of holes.

$$I_p = Aqpv_{p\text{-drift}} \quad I_p = Aqp\mu_p E \quad \text{(Coulombs)}$$

- where *A* is the cross-sectional area of the silicon bar and *q* is the magnitude of electron charge.
- And the density is : $J_p = I_p/A = qp\mu_p E$
- electrons drifting from right to left result in a current component from left to right.

$$I_n = -Aqnv_{n\text{-drift}}$$

- and

$$J_n = I_n/A \quad = \quad J_n = qn\mu_n E$$

- The total drift current density can now be found by summing *Jp* and *Jn*

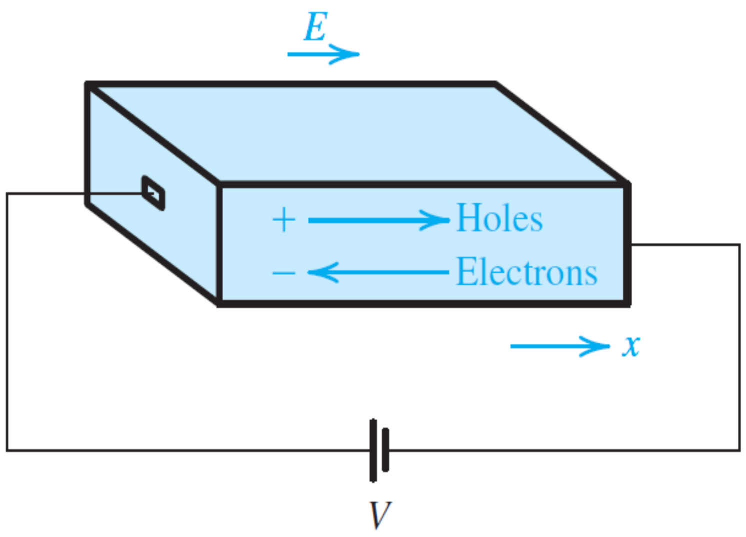
$$J = J_p + J_n = q(p\mu_p + n\mu_n)E$$

This relationship can be written as

$$J = \sigma E$$

Current Flow in Semiconductors

Drift Current



or

$$J = E/\rho$$

where the **conductivity** σ is given by

$$\sigma = q(p\mu_p + n\mu_n)$$

and the **resistivity** ρ is given by

$$\rho \equiv \frac{1}{\sigma} = \frac{1}{q(p\mu_p + n\mu_n)}$$

And Ohm’s law can be written alternately as

$$\rho = \frac{E}{J}$$

Thus the units of ρ are obtained from: $\dfrac{\text{V/cm}}{\text{A/cm}^2} = \Omega \cdot \text{cm}.$

Current Flow in Semiconductors

Example 3

Find the resistivity of (a) intrinsic silicon and (b) p-type silicon with $N_A = 10^{16}/\text{cm}^3$. Use $n_i = 1.5 \times 10^{10}/\text{cm}^3$, and assume that for intrinsic silicon $\mu_n = 1350\ \text{cm}^2/\text{V}\cdot\text{s}$ and $\mu_p = 480\ \text{cm}^2/\text{V}\cdot\text{s}$, and for the doped silicon $\mu_n = 1110\ \text{cm}^2/\text{V}\cdot\text{s}$ and $\mu_p = 400\ \text{cm}^2/\text{V}\cdot\text{s}$. (Note that doping results in reduced carrier mobilities.)

Solution

(a) For intrinsic silicon,

$$p = n = n_i = 1.5 \times 10^{10}/\text{cm}^3$$

$$\rho = \frac{1}{q(p\mu_p + n\mu_n)}$$

$$\rho = \frac{1}{1.6 \times 10^{-19}(1.5 \times 10^{10} \times 480 + 1.5 \times 10^{10} \times 1350)}$$

$$= 2.28 \times 10^5\ \Omega\cdot\text{cm}$$

Current Flow in Semiconductors

Example 3

Solution

(b) For the p-type silicon

$$p_p \simeq N_A = 10^{16}/\text{cm}^3$$

$$n_p \simeq \frac{n_i^2}{N_A} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4/\text{cm}^3$$

Thus,

$$\rho = \frac{1}{q(p\mu_p + n\mu_n)}$$

$$= \frac{1}{1.6 \times 10^{-19}(10^{16} \times 400 + 2.25 \times 10^4 \times 1110)}$$

$$\simeq \frac{1}{1.6 \times 10^{-19} \times 10^{16} \times 400} = 1.56\ \Omega \cdot \text{cm}$$

Current Flow in Semiconductors/Drift current

EXERCISE 1

A uniform bar of n-type silicon of 2-μm length has a voltage of 1 V applied across it. If $N_D = 10^{16}/\text{cm}^3$ and $\mu_n = 1350\ \text{cm}^2/\text{V}\cdot\text{s}$, find (a) the electron drift velocity, (b) the time it takes an electron to cross the 2-μm length, (c) the drift-current density, and (d) the drift current in the case that the silicon bar has a cross-sectional area of 0.25 μm^2.

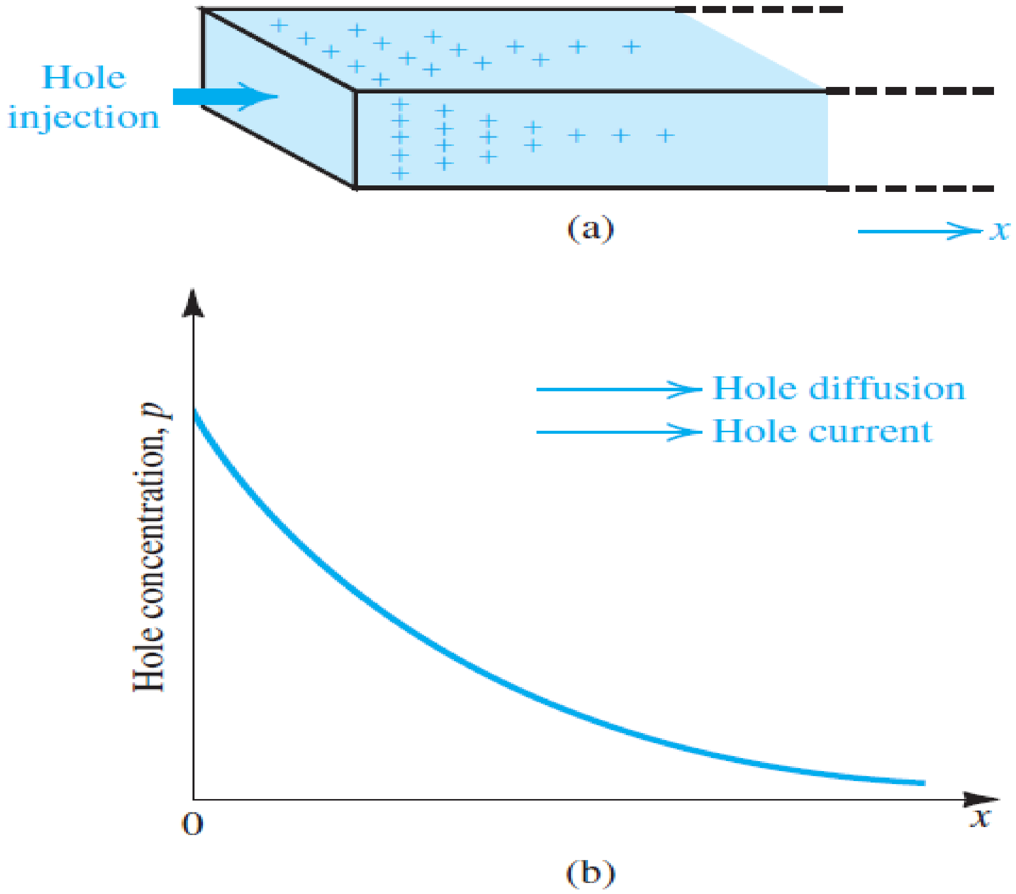
Current Flow in Semiconductors

Diffusion Current

Carrier diffusion occurs when the density of charge carriers in a piece of semiconductor is not uniform. the concentration of, say, holes, is made higher in one part of a piece of silicon than in another,

→ holes will diffuse from the region of high concentration to the region of low concentration (fig a)

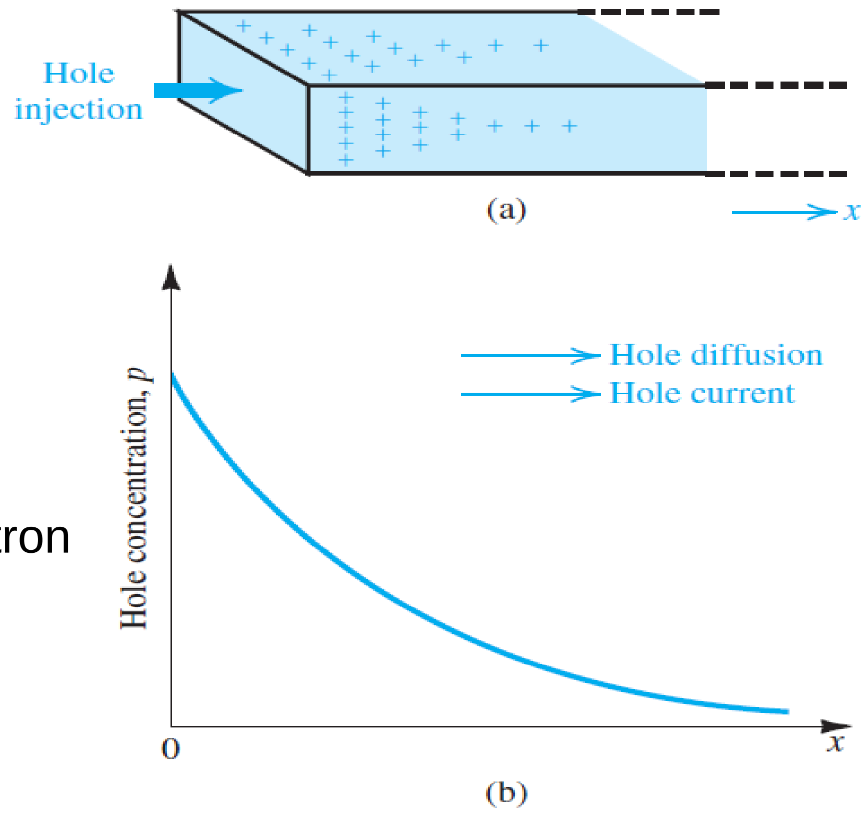
→ In fig b) The holes diffuse in the positive direction of *x* and give rise to a hole diffusion current in the same direction.



A bar of silicon **(a)** into which holes are injected, **(b)** the hole concentration profile along the *x* axis.

Physics of Semi-Conductors/Current Flow in Semiconductors

Diffusion Current



- The magnitude of the current at any point is proportional to the slope of the concentration profile, or the **concentration gradient**, at that point,

$$J_p = -qD_p \frac{dp(x)}{dx}$$

- ✓ where J_p is the hole-current density (A/cm^2), q is the magnitude of electron charge,
- ✓ D_p is a constant called the **diffusion constant** or **diffusivity** of holes;
- ✓ and $p(x)$ is the hole concentration at point x.

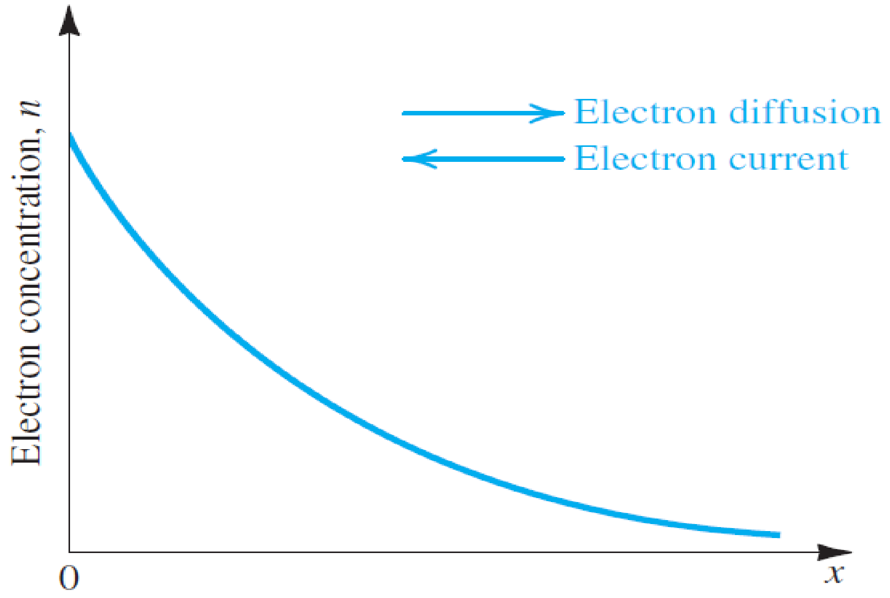
the gradient (dp/dx) is negative, resulting in a positive current in the x direction

Physics of Semi-Conductors/Current Flow in Semiconductors

Diffusion Current

In the case of electron diffusion resulting from an electron concentration gradient, a similar relationship applies, giving the electron-current density,

$$J_n = qD_n \frac{dn(x)}{dx}$$



where *Dn* is the diffusion constant or diffusivity of electrons.

For holes and electrons diffusing in intrinsic silicon, typical values for the diffusion constants are D_p = 12 cm/s and D_n = 35 cm$_2$/s.

If the electron concentration profile shown is established in a bar of silicon, electrons diffuse in the *x* direction, giving rise to an electron diffusion current in the negative-*x* direction.

Example 4

Consider a bar of silicon in which a hole concentration profile described by

$$p(x) = p_0 e^{-x/L_p}$$

is established. Find the hole-current density at $x = 0$. Let $p_0 = 10^{16}/\text{cm}^3$, $L_p = 1\ \mu\text{m}$, and $D_p = 12\ \text{cm}^2/\text{s}$. If the cross-sectional area of the bar is $100\ \mu\text{m}^2$, find the current I_p.

Solution

$$\begin{aligned} J_p &= -qD_p \frac{dp(x)}{dx} \\ &= -qD_p \frac{d}{dx}\left[p_0 e^{-x/L_p}\right] \\ &= q\frac{D_p}{L_p} p_0 e^{-x/L_p} \end{aligned}$$

Physics of Semi-Conductors/Current Flow in Semiconductors

Example 4

Solution

Thus,

$$J_p(0) = q\frac{D_p}{L_p}p_0$$

$$= 1.6 \times 10^{-19} \times \frac{12}{1 \times 10^{-4}} \times 10^{16}$$

$$= 192\ \mathrm{A/cm^2}$$

The current I_p can be found from

$$I_p = J_p \times A$$

$$= 192 \times 100 \times 10^{-8}$$

$$= 192\ \mu\mathrm{A}$$

Physics of Semi-Conductors/Current Flow in Semiconductors

EXERCISE 2

The linear electron-concentration profile shown in Fig. 1, has been established in a piece of silicon. If $n_0 = 10^{17}/\text{cm}^3$ and $W = 1\ \mu\text{m}$, find the electron-current density in microamperes per micron squared ($\mu\text{A}/\mu\text{m}^2$). If a diffusion current of 1 mA is required, what must the cross-sectional area (in a direction perpendicular to the page) be? Recall that $D_n = 35\ \text{cm}^2/\text{s}$.

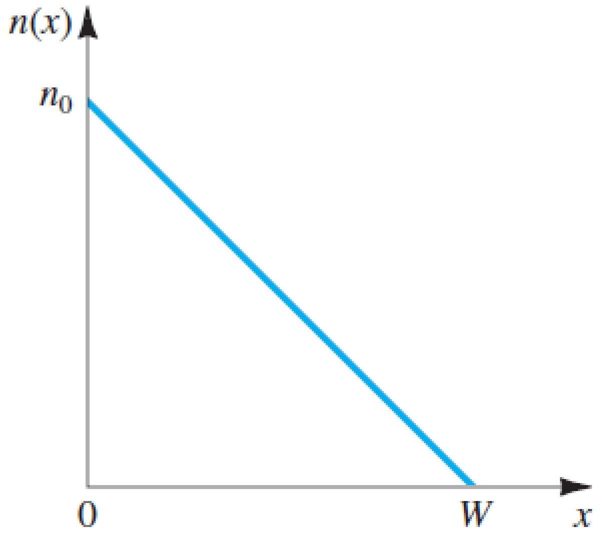


Fig. 1